

Jerry Liu

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Interests: Fundamental ML architectures & algorithms; in-context learning & algorithmic reasoning; ML for numerical & scientific computation.

EDUCATION

Stanford University — Ph.D. Candidate, Computational & Mathematical Engineering Sep 2022 — Winter 2026 (expected)

Advisor: Chris Ré @ Hazy Lab

- **GPA:** 4.03. DOE Computational Science Graduate Fellowship.

Duke University — B.S. Mathematics & Computer Science, Minor in Music

Aug 2018 — Dec 2021

- **GPA:** 3.99; Summa Cum Laude; Phi Beta Kappa.

WORK EXPERIENCE

Lawrence Berkeley National Lab, Machine Learning & Analytics — Research Intern

Jun 2023 — Sep 2023

Advisors: N. Benjamin Erichson, Michael W. Mahoney

Berkeley, CA

- Analyzed transformer architectures for **algorithmic reasoning in differential equation systems**, focusing on in-context system identification.
- Established theoretical and empirical connections between **attention mechanisms and pseudospectral PDE solvers** and evaluated generalization under distribution shifts

Duolingo, Learning AI Lab — Software Engineer

Jan 2022 — Sep 2022

Pittsburgh, PA

- Enhanced user knowledge ML model serving **9 million DAUs** with multilingual LLMs
- Implemented tools for model performance visualization; productionized data cleaning and model training pipelines

Lawrence Livermore National Lab — Research Intern

Jun 2021 — Dec 2021

Advisors: Jin Yao, Kenneth Weiss

Livermore, CA

- Developed interface reconstruction algorithms with geometric optimization schemes to solve incompressible Navier-Stokes equations
- Formulated and implemented numerical schemes for advection with hybrid reconstruction-tracking method for multi-material fluid flow problems

SELECTED PUBLICATIONS

- Owen Dugan*, Roberto Garcia*, Ronny Junkins*, **Jerry Liu***, Dylan Zinsley, Sabri Eyuboglu, Atri Rudra, Chris Ré. *Constructing Efficient Fact-Storing MLPs for Transformers*. In submission. [[paper](#) | [blog](#)]
- **Jerry Liu**, Yasa Baig, Denise Hui Jean Lee, Rajat Vaddiraj Dwaraknath, Atri Rudra, Chris Ré. *BWLER: Barycentric Weight Layer Elucidates a Precision-Conditioning Tradeoff for PINNs*. TASC @ COLT 2025 (**Best Paper Award**) [[paper](#) | [code](#) | [blog](#)]
- **Jerry Liu**, Jessica Grogan, Owen Dugan, Ashish Rao, Simran Arora, Atri Rudra, Chris Ré. *Towards Learning High-Precision Least Squares Algorithms with Sequence Models*. ICLR 2025. [[paper](#) | [code](#)]
- Christopher Fifty, Ronald Junkins, Dennis Duan, Aniketh Iyengar, **Jerry Liu**, Ehsan Amid, Sebastian Thrun, Chris Ré. *Restructuring Vector Quantization with the Rotation Trick*. ICLR 2025 **Oral (top 1.8%)**. [[paper](#) | [code](#)]
- **Jerry Liu**, N. Benjamin Erichson, Kush Bhatia, Michael W. Mahoney, Chris Ré. *Does In-Context Operator Learning Generalize to Domain-Shifted Settings?* DLDE III, NeurIPS 2023. [[paper](#)]

HONORS

- **Department of Energy — Computational Science Graduate Fellow** 2022
- **NSF GRFP (declined)** 2022
- **CRA Outstanding Undergraduate Researcher — Honorable Mention** 2022
- **William Lowell Putnam Mathematical Competition – top 250** 2018 — 2019

SKILLS

Machine Learning PyTorch, JAX; model training and evaluation workflows; transformers and long-context architectures
Mathematics Numerical linear algebra, optimization, numerical PDEs, statistical learning theory
Programming Python, C++, Java, MATLAB, SQL